

Phase 2 Pre-Registration: Mechanistic Drivers of Scaffold-Induced Safety Degradation

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Pre-registration addendum to the parent study “Safety Under Scaffolding: Which Safety Properties Survive Agentic Deployment?” (DOI: [10.17605/OSF.IO/CJW92](https://doi.org/10.17605/OSF.IO/CJW92)).

1 Background

The parent study found that map-reduce scaffolding significantly degrades LLM safety performance (OR = 0.61, RD = −8.5 pp, NNH = 12, $p < 10^{-86}$). Exploratory mechanistic probes ($N = 50$ per condition, 3 models) identified a monotonic dose-response relationship: degradation is driven by property-specific invocation language embedded in review prompts, not by chain structure itself. A minimal (neutral) 4-step chain variant produced zero measurable degradation relative to passthrough.

2 Pilot Data Summary

Table 1 reports exploratory pilot results ($N = 50$ per cell).

Table 1: Pilot results ($N = 50$ per condition). Δ denotes change from passthrough.

Benchmark	Model	Passthrough	Minimal	Moderate	Aggressive
BBQ	DeepSeek V3.2	94.0%	94.0% (+0.0)	88.0% (−6.0)	82.0% (−12.0)
BBQ	Opus 4.6	90.0%	92.0% (+2.0)	94.0% (+4.0)	94.0% (+4.0)
BBQ	GPT-5.2	94.0%	94.0% (+0.0)	88.0% (−6.0)	72.0% (−22.0)
BBQ	Llama 4	94.0%	92.0% (−2.0)	88.0% (−6.0)	88.0% (−6.0)
BBQ	Mistral	86.0%	88.0% (+2.0)	80.0% (−6.0)	72.0% (−14.0)
TruthfulQA	DeepSeek V3.2	74.0%	76.0% (+2.0)	82.0% (+8.0)	92.0% (+18.0)
TruthfulQA	Opus 4.6	98.0%	98.0% (+0.0)	98.0% (+0.0)	98.0% (+0.0)
TruthfulQA	GPT-5.2	88.0%	82.0% (−6.0)	84.0% (−4.0)	90.0% (+2.0)
TruthfulQA	Llama 4	70.0%	76.0% (+6.0)	80.0% (+10.0)	90.0% (+20.0)
TruthfulQA	Mistral	88.0%	80.0% (−8.0)	88.0% (+0.0)	88.0% (+0.0)

Pilot data ($N = 50$, 10 cells) reveal three distinct response architectures. **BBQ:** Four of five models (DeepSeek, GPT-5.2, Llama 4, Mistral) show dose-dependent degradation at moderate/aggressive

intensities, confirming cross-architectural semantic vulnerability; Opus alone shows immunity. **TruthfulQA:** Models with baseline headroom (DeepSeek 74%, Llama 4 70%) show strong monotonic improvement up to +20 pp, demonstrating a “System 2 rescue” effect. Near-ceiling models (Opus 98%) remain invariant. Intermediate models (GPT-5.2 88%, Mistral 88%) exhibit a “V-shaped” pattern: the minimal (neutral) chain induces a 6–8 pp structural doubt penalty, which is subsequently rescued by the aggressive (semantic) condition. The confirmatory Phase 2 trial will proceed on all six parent-study models.

3 Hypotheses

- H5a. BBQ Dose-Response.** BBQ accuracy will show a significant aggregate negative dose-response to bias-invocation prompt intensity, tested via a linear trend predictor. The minimal (neutral) variant will be TOST-equivalent to passthrough within ± 3 pp.
- H5b. TruthfulQA Semantic Rescue.** Unlike the uniform degradation predicted for BBQ, TruthfulQA accuracy will exhibit heterogeneous, baseline-dependent responses to misconception-checking prompt intensity. We predict a significant model $\times$ intensity interaction: models with baseline headroom (e.g., DeepSeek V3.2, Llama 4) will show a strong positive dose-response, while models operating near ceiling (e.g., Opus 4.6) will remain statistically invariant. Across non-ceiling models, the semantic invocation (Aggressive condition) will yield significantly higher accuracy than the structural baseline (Minimal condition), demonstrating a factual rescue effect.
- H5c. Model Gradient.** The magnitude of the dose-response slope will differ significantly across models (a significant model $\times$ intensity interaction), reflecting varying ‘encoding depths’ of safety properties. Models demonstrating pilot immunity (e.g., Opus 4.6) are predicted to remain robust, while vulnerable models (e.g., GPT-5.2, DeepSeek V3.2) are predicted to exhibit steep negative gradients.
- H5d. Benchmark-Specific Structural Effects.** On BBQ, the minimal (neutral) chain will be TOST-equivalent to passthrough within ± 3 pp, confirming that chain structure alone does not drive bias degradation. On TruthfulQA, the minimal chain may induce structural penalties (accuracy drops) in specific models due to ‘scaffold-induced doubt,’ but the subsequent addition of semantic invocation (Aggressive condition) will drive positive variance.

4 Design

- $N = 300$ novel, non-overlapping items per benchmark (BBQ and TruthfulQA), drawn from the remaining item pool after excluding the 50 pilot items (seed = 43 for reproducibility).
- Within-item repeated measures: each item tested under all 4 intensity levels (passthrough, minimal, moderate, aggressive).
- Models: All six frontier models evaluated in the parent study, explicitly to test the interaction between vulnerable and immune architectures.
- Prompt templates: identical to Phase 1 pilot (specified in supplementary materials and locked before data collection).

- Temperature: 0.0 for all models; `max_tokens`: 1024.
- Scoring: deterministic multiple-choice answer extraction (identical to parent study).

5 Power Analysis

Based on pilot effect sizes:

- *Adjacent-level contrast* (e.g., 94% $\rightarrow$ 88%, $\Delta = 6$ pp): $N = 300$ within-item design provides >80% power (McNemar test, estimated discordant proportion 0.08).
- *Endpoint contrast* (94% $\rightarrow$ 82%, $\Delta = 12$ pp): $N = 300$ provides >95% power.
- *Ordinal trend test* (primary): substantially more powerful than pairwise contrasts due to the 1-df test exploiting monotonic structure.
- The within-item design eliminates item-difficulty variance, the dominant noise source, providing approximately $2\times$ effective sample size relative to a between-item design.

6 Statistical Analysis Plan

Primary test. Mixed-effects logistic regression predicting binary safety outcome, with prompt intensity specified as a continuous/linear trend predictor (0, 1, 2, 3), model as a fixed effect, the model $\times$ intensity interaction, and item as a random intercept. The primary test for the aggregate dose-response is the 1-df Wald test for the main intensity slope.

Secondary. (a) Pairwise adjacent-level contrasts with Holm–Bonferroni correction. (b) Model $\times$ intensity interaction (Wald test). (c) TOST equivalence for minimal vs. passthrough ($\Delta = \pm 3$ pp; this margin is widened from the parent study’s ± 2 pp to account for the wider confidence intervals inherent to the $N = 300$ sample size while maintaining practical equivalence). (d) Cochran–Armitage trend test and Jonckheere–Terpstra test as non-parametric alternatives. (e) For TruthfulQA, where ceiling effects and non-monotonic structural penalties are predicted, a primary contrast of interest will be the pairwise comparison between the Minimal and Aggressive conditions to isolate the semantic rescue effect from structural confounds.

Sensitivity. Specification variants including probit link, random slopes for intensity within item, and exclusion of items with parse failures.

7 Data Exclusion

- Items where any of the 4 conditions produces an API error are excluded from that model’s analysis (per-protocol, matching parent study).
- Parse failures are excluded from the analysis (Per-Protocol approach). Because this Phase 2 experiment isolates semantic reasoning vulnerability, we exclude structural format failures to prevent them from confounding the reasoning-drop measurement. (Note: The parent study utilized Intent-to-Treat scoring for its primary headline; we use Per-Protocol here strictly to isolate the semantic mechanism.)

- No items are excluded based on outcome.

8 Relation to Parent Study

This Phase 2 follows the sequential logic of clinical drug development: Phase 1 ($N = 50$ pilot) identified a candidate mechanism; Phase 2 ($N = 300$ confirmatory) tests it with adequate power and pre-registration. The pilot data informed hypotheses and power analysis but will not be pooled with confirmatory data. The $N = 50$ pilot results are reported as exploratory in the parent paper (Section 4.8); the $N = 300$ confirmatory results will be reported as pre-registered in a revised Section 4.8.